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Digital Material Passport

ID	DMP-2024-ARR-001	Version	0.1.0
Issue Date	2024-12-15	Certificate Type	EN 10204 3.1

Business Transaction

Order		Delivery	
Order ID	PO-2024-7890	Delivery ID	DN-2024-3456
Quantity		Quantity	500 kg

Product Information

Product Name	Hardenability Test Steel
Batch ID	HTB-2024-045

Product Shape

Form	RoundBar
Length	1000 mm
Diameter	100 mm

Chemical Analysis

Heat Number	H-2024-078
Sample Location	Ladle

Elements

Symbol	C	Mn	Si	P	S	Cr	Ni
Unit	%	%	%	%	%	%	%
Min	0.42	0.5	-	-	-	-	-
Max	0.5	0.8	0.4	0.03	0.025	0.25	0.25
Actual	0.45	0.65	0.25	0.015	0.008	0.12	0.08

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status				
Jominy Hardenability Test Quenched from 850°C					ISO 377	✓				
Distance from - quenched end (mm)	1.5	3	5	7	9	11	15	20	25	30
Value [HRC]	45	43	40	36	33	31	27	25	28	27
Min	42	39	35	32	29	26	22	20	-	-
Max	47	46	44	41	39	37	33	31	<=30	<=29
Status	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Coating Thickness Profile Magnetic induction method					ISO 2178	✓
Position along length (mm)	0		50		100	
Value [µm]	125		115		108	
Min	>=120		>=110		>=100	
Max	<=130		<=125		<=120	
Status	Pass		Pass		Pass	

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated By

Name	Title	Department	Date
Dr. Maria Schmidt	Metallurgical Engineer	Quality Control Laboratory	2024-12-15